



How News Recommender Systems Work — and What Recent Models Improve

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❖ CROWN

❖ LANCER

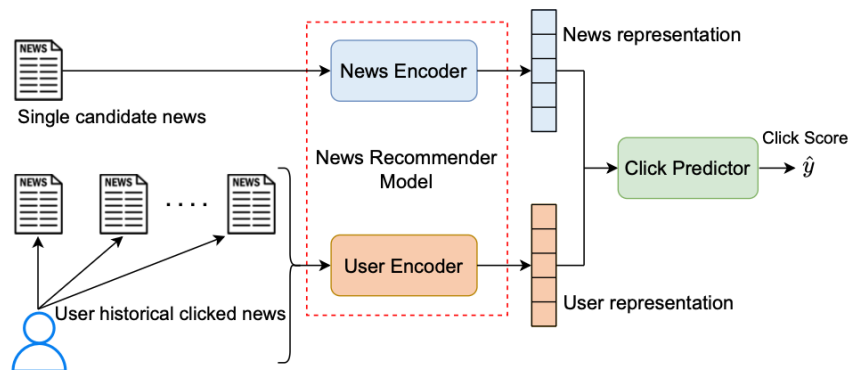
❖ LIME

News Recommendation

❖ Goal of News Recommendation

- ❑ Predict which news articles that the user is most likely to read

❖ Typical Approach

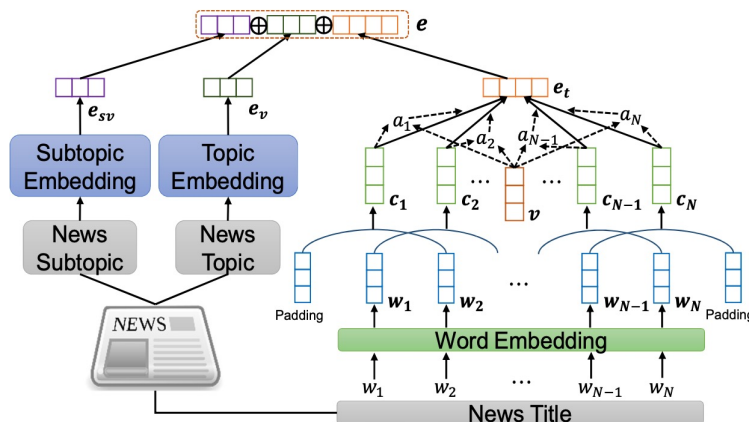


- ❑ Model the **user's interests** based on the news that user has clicked on
- ❑ Recommend the **candidate news** that is most similar to the modeled user interest

News Encoder

❖ Representing News with Rich Item Information

- ❑ Text-based Information
 - **Title**, Abstract, Body
- ❑ Metadata
 - Time-based features (age, lifetime)
- ❑ Category information
 - **Topic**, Subtopic



[1] Title, Topic, Subtopic 활용한 News Encoder

❖ Importance of News Embeddings

- ❑ Serve as core inputs for building the **user embedding**
- ❑ Used with **candidate news embeddings** to compute the final relevance score

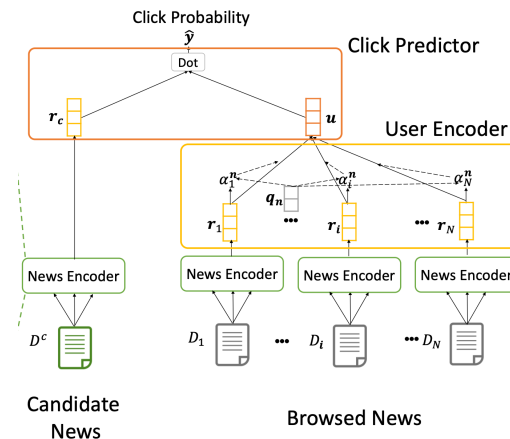
User Encoder & Click Predictor

❖ Building User Embeddings from News Embeddings

- ❑ Use **attention** to assign importance weights to each clicked news embedding
- ❑ **Aggregate weighted news embeddings** to form the final user embedding

❖ Estimating Click Probability

- ❑ Candidate news embeddings are generated by the same News Encoder
- ❑ Compute the **click probability** by comparing the user embedding with each candidate news embedding



[1] User Encoder and Click Predictor

Dataset (MIND)

❖ How Data for News Recommendation Is Structured

❑ User behaviors data

▸ Impressions

- Among the news shown to a user at a given time, clicked items are labeled 1 and non-clicked items 0

▸ History

- IDs of news articles the user has clicked up to that moment
- Category, SubCategory

❑ Text-based fields such as title and abstract

Column	Content
Impression ID	91
User ID	U397059
Time	11/15/2019 10:22:32 AM
History	N106403 N71977 N97080 N102132 N97212 N121652
Impressions	N129416-0 N26703-1 N120089-1 N53018-0 N89764-0 N91737-0 N29160-0

Column	Content
News ID	N37378
Category	sports
SubCategory	golf
Title	PGA Tour winners
Abstract	A gallery of recent winners on the PGA Tour.
URL	https://www.msn.com/en-us/sports/golf/pga-tour-winners/ss-AAjnQjj?ocid=chopendata
Title Entities	[{"Label": "PGA Tour", "Type": "O", "WikidataId": "Q910409", "Confidence": 1.0, "OccurrenceOffsets": [0], "SurfaceForms": ["PGA Tour"]}]
Abstract Entities	[{"Label": "PGA Tour", "Type": "O", "WikidataId": "Q910409", "Confidence": 1.0, "OccurrenceOffsets": [35], "SurfaceForms": ["PGA Tour"]}]

Key Challenges in News Recommendation

❖ Characteristics of News Items

❑ Highly **time-sensitive**

- ▶ Consumption quickly spikes right after publication and interest fades rapidly
- ▶ Shows a very different consumption pattern compared to items like movies

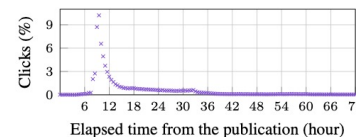


Figure 1: Distribution of average click ratios for news by users over time (Adressa).

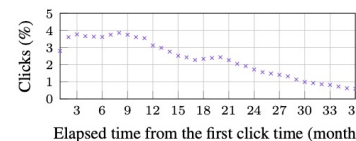


Figure 2: Distribution of average click ratios for items by users over time (Netflix).

❖ User Interests are Diverse

- ❑ Single user can have preferences across multiple categories
- ❑ Both long-term preferences and short-term
 - ▶ e.g., sports news during the Olympics
 - ▶ e.g., political news during an election season

[1] Overall click rate time distribution for News and Movie

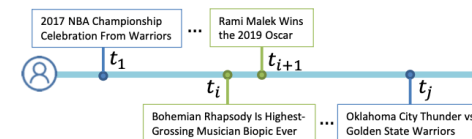


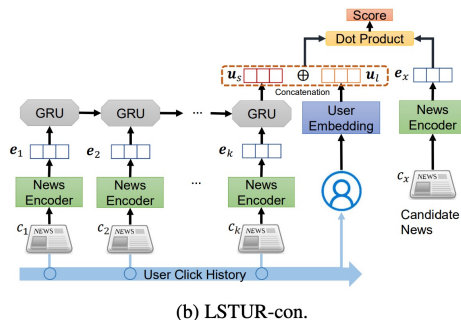
Figure 1: An illustrative example of long-term and short-term interests in news reading.

[2] Long-term, short-term interests

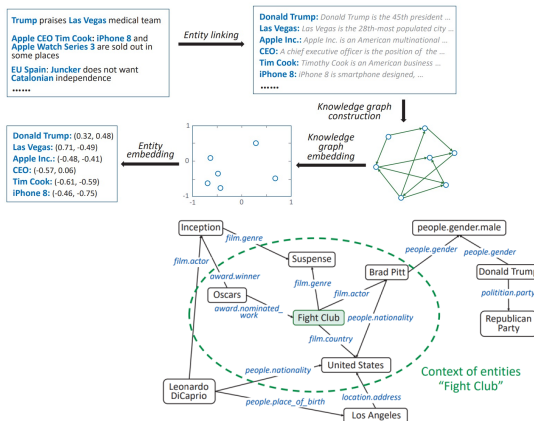
[1] Bae, H.-K., Ahn, J., Lee, D., & Kim, S.-W. (2023). LANCER: A Lifetime-Aware News Recommender System. Proceedings of the AAAI Conference on Artificial Intelligence, 37(4), 4141-4148.

[2] Mingxiao An, Fangzhao Wu, Chuhan Wu, Kun Zhang, Zheng Liu, and Xing Xie. 2019. Neural News Recommendation with Long- and Short-term User Representations. In Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics, pages 336-345, Florence, Italy. Association for Computational Linguistics.

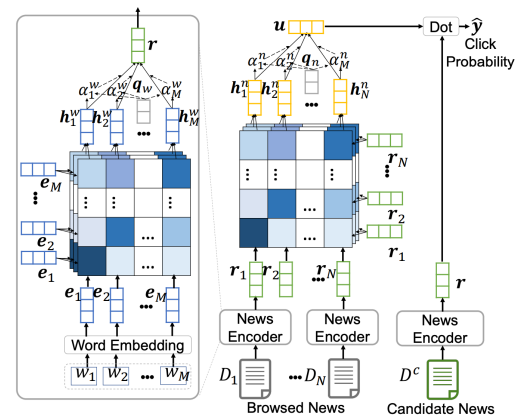
Previous Approaches



Considering User's Long, Short-term
(LSTUR, ACL 19')



Matching title words with external KG
entities(DKN, WWW 18')



Utilizes multi-head self-attention
(NRMS, EMNLP 19')

News Encoder + User Encoder + Click Predictor

- Previous work focuses on generating expressive embeddings through improved news and user encoders

Studies also explore fairness, sentiment, privacy, and more

Upcoming Topics

❖ More Refined User Behavior Modeling

- ❑ Do all user's news clicks carry meaningful signals?
- ❑ If some clicks are just noise, how can we detect and filter them?

❖ More Granular News Representation

- ❑ Is there a more fine-grained way to classify news beyond categories?

❖ New Features for Time-Sensitive Items

- ❑ Are older news articles less likely to be selected as time passes?
- ❑ Does this tendency differ across news categories?



CROWN: A Novel Approach to Comprehending Users' Preferences for Accurate Personalized News Recommendation

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Chung-Ang University

Seoul, Korea

Seongeun Ryu

Hanyang University

Seoul, Korea

Sang-Wook Kim

Hanyang University

Seoul, Korea

CROWN (WWW 25')

❖ Distinguishing Intents Within Categories

- ❑ Each news article is created with a specific intent
- ❑ These intents can vary from one article to another

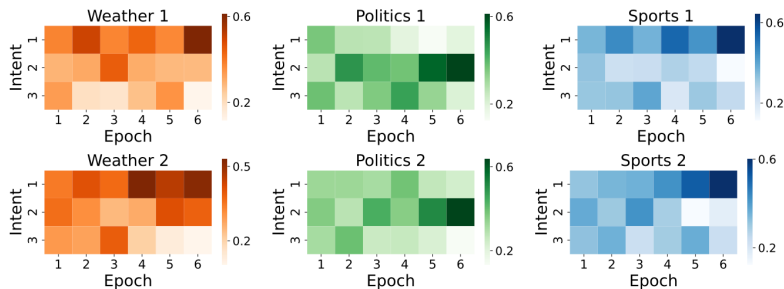
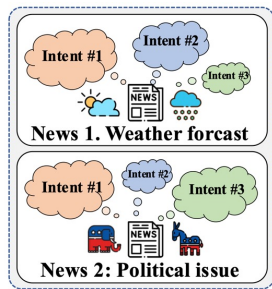
❖ Post-read Preference

- ❑ User **clicks based on the title** and the **internal content aligns** with it, may continue to prefer
- ❑ Preference strength can be adjusted based on alignment between title and content

❖ Solving the Cold-Start User Problem with GNNs

- ❑ When users have limited click history, leverage collaborative signals from another users

Comprehension of manifold intents



❖ Single News Article Can Contain Multiple Intents

☐ Weather news

- ▶ Provide information + improve public interest
- ▶ Difficult to regard them as having persuasive intent

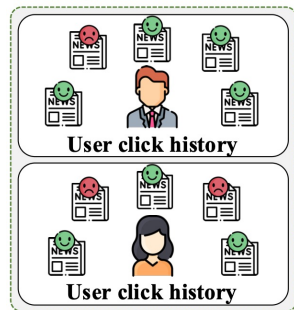
☐ Politic news

- ▶ Provide information about policy + persuade to (dis)agree
- ▶ Difficult to regard them as purely serving public interest

Post-read Preference

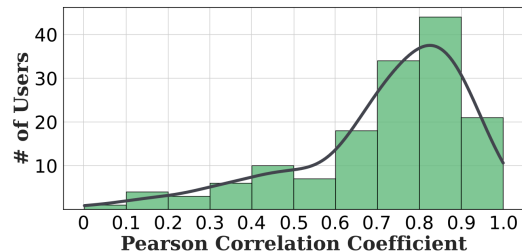
❖ Do Users Still Prefer the News Article After Clicking?

- ❑ Users decide whether to read based on the title
 - If the title and content are consistent, users tend to stay longer
 - If they are inconsistent, users often leave immediately
- ❑ A user's **post-read preference** can be estimated by how long they stay on the article



❖ Correlation Between Title–Body Consistency and Reading Time

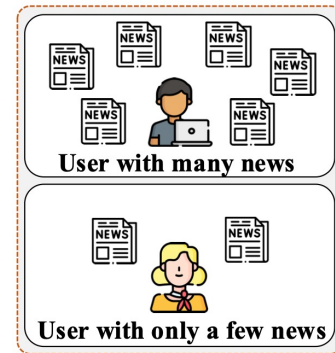
- ❑ Users stay longer when the title and body are well-aligned
- ❑ Many users show shorter reading times in opposite case



Cold-Start User Problem

❖ Difficult to make accurate recommendations under less information

- ❑ Limited click history, less information for recommendation

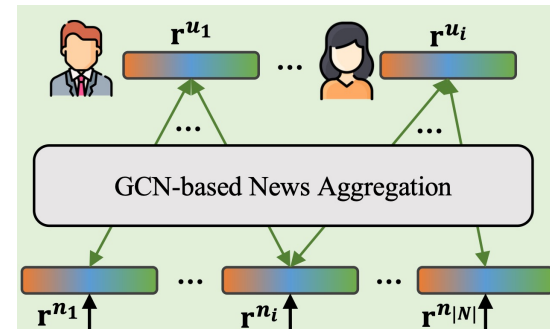


❖ Users with Similar Interests Tend to Consume Similar Items

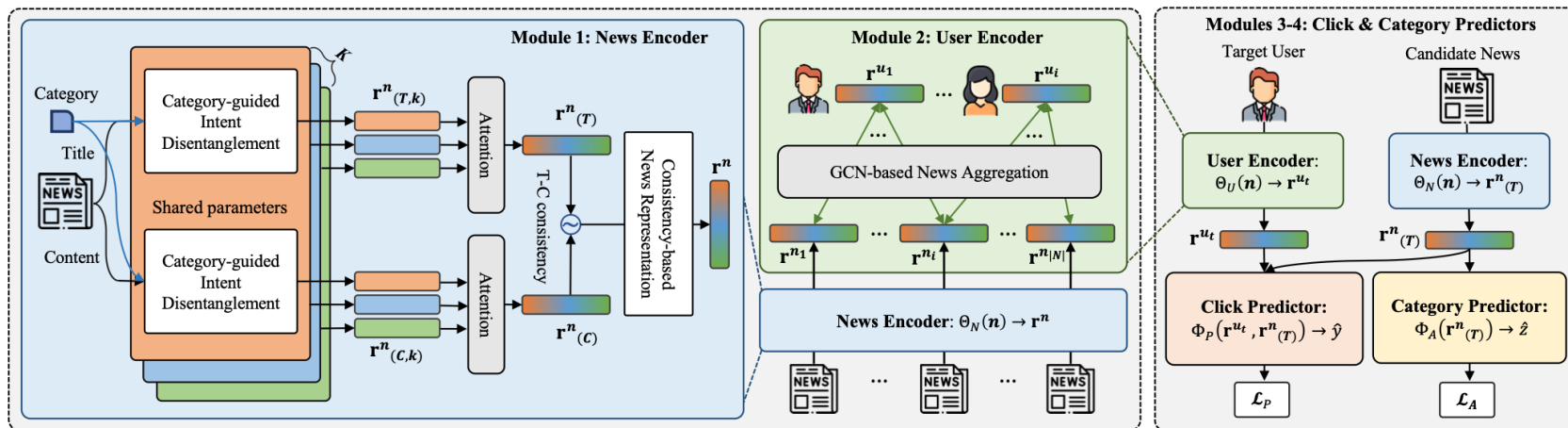
- ❑ Information from similar users can be beneficial

❖ Introducing GNNs into the User Encoder

- ❑ Content-based + Collaborative (Hybrid)
- ❑ Incorporates not only a user's own clicks but also collaborative signals from similar users



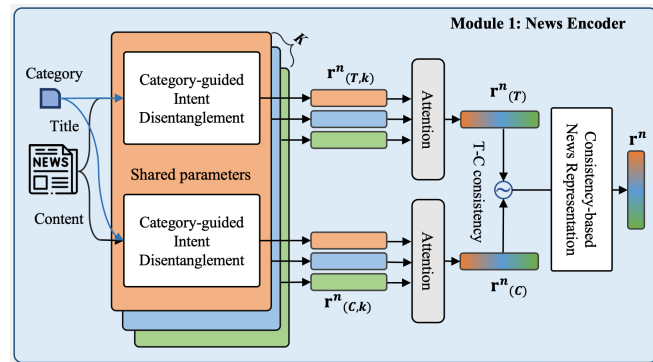
CROWN (CategoRy-guided intent disentanglement and cOnsistency-based neWs represeNtation)



News Encoder

❖ Category-guided Intent Disentanglement

- ❑ News within the **same category tend to share similar intents**
- ❑ Titles and bodies of news are represented with separate embeddings that capture different intents
 - Each embedding reflects a distinct intent
 - Learned with guidance from category information



❖ Combining Disentangled Embeddings to Form the Final News Embedding

- ❑ Title-body consistency captures meaningful user preferences
 - Incorporates this consistency in news representation
- ❑ Attention is used to weight intent-specific embeddings

Category-guided Intent Disentanglement

News Title

$$T_n = \{w_1, w_2, \dots, w_{|T_n|}\} \Rightarrow \mathbf{r}_T^n = \text{Avg}(\text{MAB}(T_n, T_n))$$

News Content

$$C_n = \{w_1, w_2, \dots, w_{|C_n|}\} \Rightarrow \mathbf{r}_C^n = \text{Avg}(\text{MAB}(C_n, C_n))$$

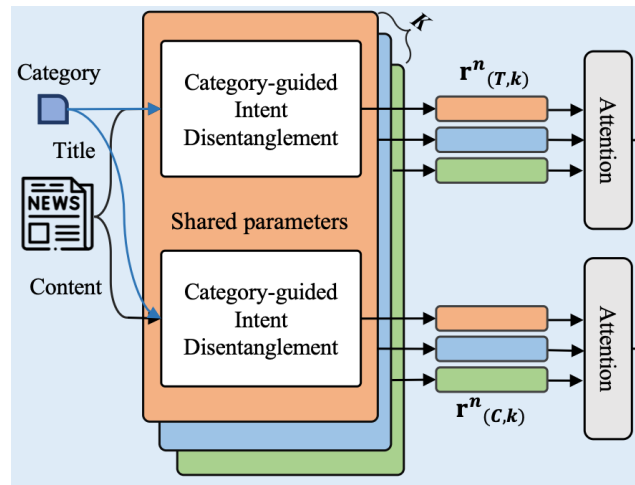
Category, Subcategory

$$\mathbf{c}_*^n = \text{Mix}(c_n, sc_n) = (c_n \oplus sc_n) \cdot \mathbf{W}_{mix} + \mathbf{b}_{mix}$$

$$\mathbf{r}_{(T,k)}^n = \text{DisEntangle}(\mathbf{r}_T^n, \mathbf{c}_*^n) = \sigma^{ReLU}((\mathbf{r}_T^n \oplus \mathbf{c}_*^n) \cdot \mathbf{W}_k + \mathbf{b}_k),$$

$$\mathbf{r}_{(C,k)}^n = \text{DisEntangle}(\mathbf{r}_C^n, \mathbf{c}_*^n) = \sigma((\mathbf{r}_C^n \oplus \mathbf{c}_*^n) \cdot \mathbf{W}_k + \mathbf{b}_k)$$

$$\mathbf{r}_{(T)}^n = \sum_{k=1}^K \alpha_{(T,k)} \cdot \mathbf{r}_{(T,k)}^n, \quad \mathbf{r}_{(C)}^n = \sum_{k=1}^K \alpha_{(C,k)} \cdot \mathbf{r}_{(C,k)}^n$$



❖ Generate “intent-aware” title/content embedding

Consistency-based Representation

❖ Computing Consistency Between Title and Body

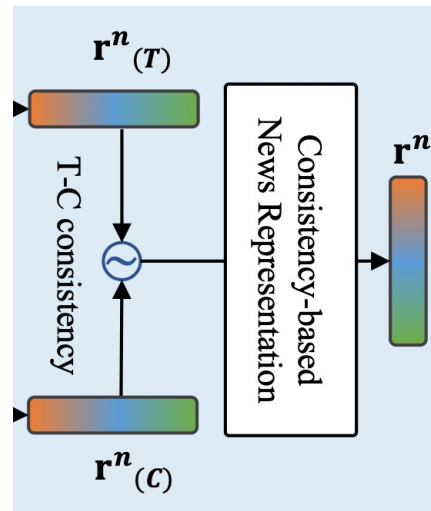
$$cs^n = \text{Consistency}(\mathbf{r}_{(T)}^n, \mathbf{r}_{(C)}^n) = \frac{\text{Cos}(\mathbf{r}_{(T)}^n, \mathbf{r}_{(C)}^n) + 1}{2}$$

- ❑ Scaled to the range [0, 1]
- ❑ **Higher values** indicate **stronger title-body alignment**
 - Interpreted as a higher likelihood that the user will be satisfied

❖ Final News Representation

$$\mathbf{r}^n = \mathbf{r}_{(T)}^n + cs^n \cdot \mathbf{r}_{(C)}^n$$

- ❑ Incorporates body embedding into the title embedding based on consistency
- ❑ When title-body alignment is strong, the body information contributes more to the final representation



User Encoder

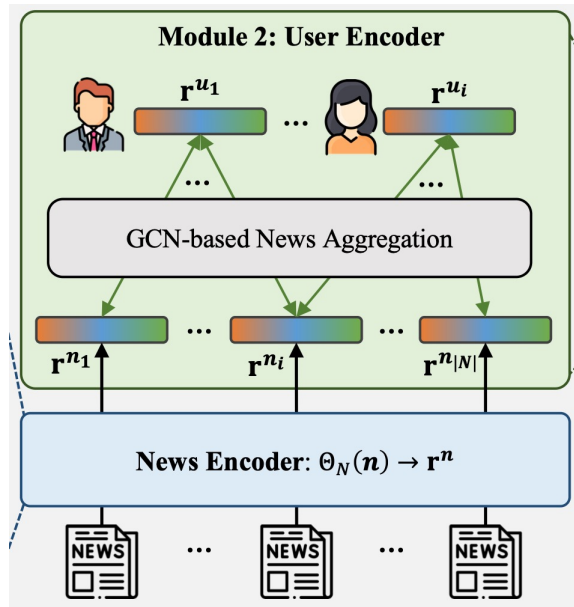
❖ Solving the Cold-Start Problem with GNNs

- ❑ Design a GNN-based hybrid user encoder
 - Apply GNNs on the user-news bipartite graph
 - Update user and news embeddings through message passing

❖ Final User Representation

$$\mathbf{r}^u = \sum_i \alpha_i \mathbf{r}_i^n$$

- ❑ Query vector used for attention weights is generated from the GNN-based user representation



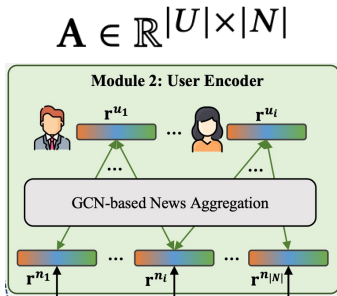
User-News Bipartite Graph

News Matrix (represented by encoder)

$$\mathbf{R}^n \in \mathbb{R}^{|N| \times d}$$

User Matrix (randomly initialized)

$$\mathbf{R}^u \in \mathbb{R}^{|U| \times d}$$



Apply GNN

$$\mathbf{R}^u = \sigma(\mathbf{A}\mathbf{R}^n\mathbf{W}_g^l + \mathbf{b}_g^l), \quad \mathbf{R}^n = \sigma(\mathbf{A}^\top \mathbf{R}^u\mathbf{W}_g^l + \mathbf{b}_g^l)$$

□ Testing GCN, GraphSAGE, LightGCN, GAT

▸ Select GAT

Table 9: The effectiveness of the GNN-enhanced user encoder on the accuracy of according to different GNN models.

		Baseline	GCN	Gain	GraphSAGE	Gain
MIND	AUC	0.6720	0.6821	+1.50%	0.6823	+1.53%
	MRR	0.3194	0.3350	+4.88%	0.3354	+5.01%
	nDCG@5	0.3550	0.3692	+4.00%	0.3697	+4.14%
	nDCG@10	0.4174	0.4304	+3.11%	0.4293	+2.85%
		Baseline	LightGCN	Gain	GAT	Gain
MIND	AUC	0.6720	0.6846	+1.88%	0.6857	+2.04%
	MRR	0.3194	0.3380	+5.82%	0.3405	+6.61%
	nDCG@5	0.3550	0.3745	+5.49%	0.3783	+6.56%
	nDCG@10	0.4174	0.4364	+4.55%	0.4384	+5.03%

Click Predictors

- ❖ Perform a dot product between user and news-title

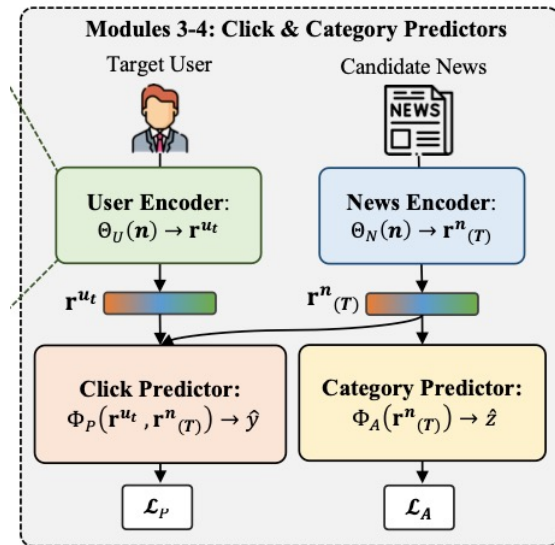
$$\hat{y}_{u,n} = \mathbf{r}^u \cdot \mathbf{r}_{(T)}^n$$

- ❑ Candidate news items are unknown about the user's click
- ❑ Content information of the candidate news is not used at this stage

- ❖ Click Prediction Loss

$$\mathcal{L}_P = - \sum_{i \in S} \log \left(\frac{\exp(\hat{y}_i^+)}{\exp(\hat{y}_i^+) + \sum_{j=1}^M \exp(\hat{y}_{i,j}^-)} \right)$$

- ❑ Trained to assign higher scores to positive samples
- ❑ Using M negative sample



Category Predictors

❖ Auxiliary Task for Guiding Better Intent Disentanglement

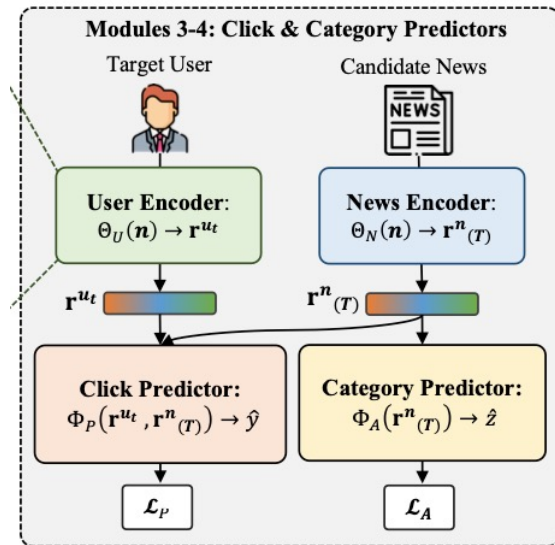
$$\mathcal{L}_A = -\frac{1}{|N|} \sum_{i=1}^{|N|} \sum_{j=1}^{|c|} z_{i,j} \cdot \log(\hat{z}_{i,j}) \quad \hat{z} = \text{softmax}(FC(\mathbf{r}_{(T)}^n))$$

- ❑ Strengthens the category-guided task
- ❑ Compare ground-truth labels with predicted probabilities

❖ Total Loss

$$\mathcal{L} = \mathcal{L}_P + \beta \mathcal{L}_A$$

- ❑ The weight of the auxiliary task is controlled by β



Experiment Setup

Table 3: Statistics of news article datasets.

Dataset	MIND-small [40]	Adressa-1week [5]
<i># of users</i>	94,057	601,215
<i># of news articles</i>	65,238	73,844
<i># of clicks</i>	347,727	2,107,312
<i># of categories</i>	18 (270)	108 (151)
<i># of words-per-title</i>	11.67	6.63
<i># of words-per-content</i>	41.01	552.15

❖ Metrics

- ❑ AUC
 - Area under the ROC curve
- ❑ MRR
 - Mean reciprocal rank
- ❑ NDCG
 - Normalized discounted cumulative gain

Rank	News	Click 여부
1	A	0
2	B	0
3	C	1 (정답)
4	D	0
5	E	1 (정답)

$$DCG = \frac{1}{\log_2(\text{rank} + 1)}$$

- DCG@10 =

$$\frac{1}{\log_2(3 + 1)} + \frac{1}{\log_2(5 + 1)} = 0.5 + 0.386 = 0.886$$

- IDCG@10 (정답이 1위, 2위였다고 가정)

$$1 + \frac{1}{\log_2(3)} = 1 + 0.6309 = 1.6309$$

- NDCG@10 =

$$0.886/1.6309 = 0.543$$

Model Accuracy

Table 4: News recommendation accuracy on two real-world datasets: CROWN consistently outperforms *all* competing methods in terms of *all* metrics (The bold font and underline indicate the best and the second-best results, respectively).

		LibFM	DSSM	NPA	NRMS	NAML	LSTUR	FIM	HieRec	CNE-SUE	DIGAT	GLORY	MCCM*	CROWN	Gain
MIND	AUC	0.6039	0.6235	0.6542	0.6623	0.6648	0.6612	0.6571	0.6718	0.6761	0.6713	0.6792	<u>0.6795</u>	0.6857	+0.91%
	MRR	0.2835	0.2934	0.3092	0.3134	0.3142	0.3121	0.3093	0.3169	0.3219	0.3225	0.3221	<u>0.3276</u>	0.3405	+3.94%
	nDCG@5	0.3040	0.3094	0.3408	0.3453	0.3480	0.3461	0.3428	0.3532	0.3572	0.3584	0.3601	<u>0.3662</u>	0.3783	+3.30%
	nDCG@10	0.3650	0.3748	0.4039	0.4135	0.4112	0.4056	0.4047	0.4161	0.4203	0.4195	0.4196	<u>0.4266</u>	0.4384	+2.77%
Adressa	AUC	0.6105	0.6682	0.6704	0.7067	0.7612	0.7673	0.7324	0.7867	<u>0.8074</u>	0.7385	0.7638	-	0.8460	+4.78%
	MRR	0.3357	0.3657	0.3885	0.4168	0.4326	0.5034	0.4252	0.4922	<u>0.5073</u>	0.4361	0.4653	-	0.5667	+11.71%
	nDCG@5	0.3083	0.3803	0.3955	0.4166	0.4453	0.4906	0.4318	0.4872	<u>0.5021</u>	0.4319	0.4761	-	0.5966	+18.82%
	nDCG@10	0.3760	0.4163	0.4279	0.4725	0.5133	0.5577	0.4874	0.5667	<u>0.5712</u>	0.4958	0.5418	-	0.6362	+11.38%

*For MCCM [31], there is no reported accuracy on the **Adressa** dataset.

News Representation

Table 5: Ablation study of the news encoder: the proposed components of the news encoder significantly contribute to enhancing the model accuracy of CROWN (The underline indicate the best result when the same number of the components are applied).

		Baseline	I.D.	C.I.	C.R.	I.D. C.I.	I.D. C.R.	C.I. C.R.	I.D. C.I. C.R.	Gain
MIND	AUC	0.6624	0.6725	<u>0.6731</u>	0.6684	0.6764	<u>0.6806</u>	0.6793	0.6857	+3.52%
	MRR	0.3139	<u>0.3201</u>	0.3192	0.3182	0.3224	<u>0.3282</u>	0.3246	0.3405	+8.47%
	nDCG@5	0.3484	<u>0.3546</u>	<u>0.3558</u>	0.3535	0.3567	<u>0.3651</u>	0.3611	0.3783	+8.58%
	nDCG@10	0.4095	0.4171	<u>0.4182</u>	0.4151	0.4194	<u>0.4259</u>	0.4230	0.4384	+7.06%
Adressa	AUC	0.7789	<u>0.8088</u>	0.7977	0.7829	0.8215	<u>0.8385</u>	0.8343	0.8460	+8.61%
	MRR	0.4831	<u>0.5094</u>	0.5047	0.5006	0.5200	<u>0.5284</u>	0.5277	0.5667	+17.30%
	nDCG@5	0.4904	<u>0.5266</u>	0.5199	0.4991	0.5331	<u>0.5533</u>	0.5518	0.5966	+21.66%
	nDCG@10	0.5546	<u>0.5886</u>	0.5868	0.5704	0.5913	<u>0.6035</u>	0.6032	0.6362	+14.71%

❖ Experiments on Combining the Three Components

- ❑ Intent Disentanglement, Category Information, Consistency-based Representation

❖ I.D. Played the Most Critical Role in News Recommendation

- ❑ C.I. and C.R. served as complementary components

User Encoder Study

❖ Hybrid Effectively Aggregates News Embeddings + Leverages Collaborative Signals

- ❑ Performance consistently improved regardless of which GNN model was used
 - GCN, GraphSAGE, GAT, LightGCN

❑ Four Variants of the User Encoder

- AVG: Simple average of embeddings from clicked news
- ATT: Weighted aggregation using attention
- GNN: Jointly updates user-news embeddings
- Hybrid: Proposed GNN-enhanced hybrid model

Table 6: The model accuracy of CROWN according to different user encoders.

		AVG	ATT	GNN	Hybrid	Gain
MIND	AUC	0.6720	0.6736	0.6764	0.6857	+2.04%
	MRR	0.3194	0.3236	0.3230	0.3405	+6.61%
	nDCG@5	0.3550	0.3594	0.3600	0.3783	+6.56%
	nDCG@10	0.4174	0.4208	0.4211	0.4384	+5.03%
Adressa	AUC	0.8216	0.8349	0.8257	0.8460	+2.97%
	MRR	0.5141	0.5223	0.5208	0.5667	+10.23%
	nDCG@5	0.5295	0.5448	0.5316	0.5966	+12.67%
	nDCG@10	0.5809	0.5851	0.5905	0.6362	+9.52%

Cold Start User Problem

Table 7: News recommendation accuracy for cold-start users: CROWN consistently outperforms *all* competing methods in terms of *all* metrics (The bold font and underline indicate the best and the second best results, respectively).

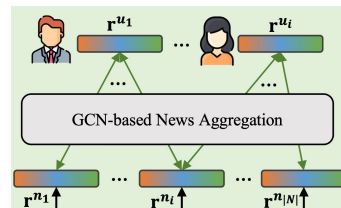
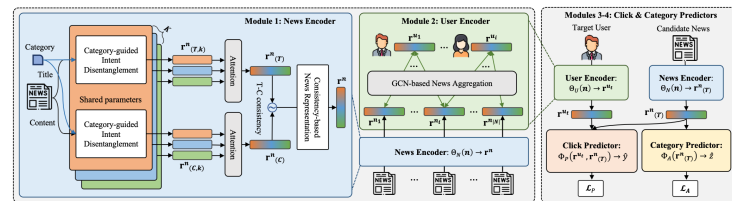
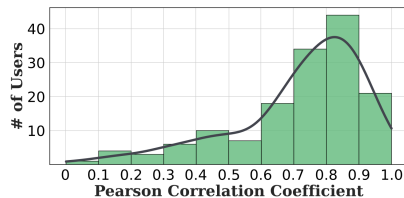
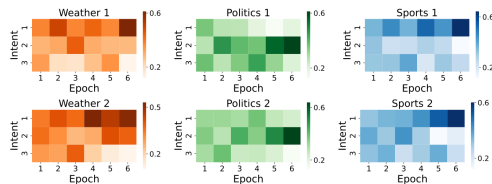
		LibFM	DSSM	NPA	NRMS	NAML	LSTUR	FIM	HieRec	CNE-SUE	DIGAT	GLORY	CROWN	Gain
MIND	AUC	0.5020	0.5832	0.6022	0.6021	<u>0.6118</u>	0.5908	0.6073	0.6104	0.5985	0.5826	0.6092	0.6352	+3.82%
	MRR	0.2509	0.2810	0.3080	0.3123	<u>0.3139</u>	0.3003	0.3066	0.3128	0.2945	0.2952	0.3081	0.3263	+3.95%
	nDCG@5	0.2592	0.3036	0.3351	0.3377	<u>0.3405</u>	0.3238	0.3389	<u>0.3425</u>	0.3275	0.3208	0.3397	0.3595	+4.96%
	nDCG@10	0.3209	0.3670	0.3951	0.3976	<u>0.4017</u>	0.3861	0.3959	<u>0.3986</u>	0.3873	0.3804	0.3964	0.4187	+4.23%
Adressa	AUC	0.5273	0.6045	0.6186	0.6351	<u>0.6718</u>	0.6787	0.6252	0.6682	<u>0.6854</u>	0.6327	0.6723	0.7236	+5.57%
	MRR	0.2863	0.3181	0.3373	0.3645	<u>0.3872</u>	0.3894	0.3436	0.3823	<u>0.4026</u>	0.3502	0.3922	0.4284	+6.41%
	nDCG@5	0.2561	0.3432	0.3529	0.3714	<u>0.3932</u>	0.4025	0.3651	0.3819	<u>0.4153</u>	0.3621	0.3948	0.4469	+7.61%
	nDCG@10	0.3358	0.3530	0.3728	0.3935	<u>0.4260</u>	0.4384	0.3885	0.4175	<u>0.4460</u>	0.3973	0.4365	0.4752	+6.55%

❖ Evaluation on Users with Fewer than 5 Clicked News

- ❑ Achieved the highest accuracy
- ❑ Performance drop was much smaller compared to existing methods
 - Demonstrates strong robustness in cold-start scenarios

Conclusion

- News articles carry different intents
 - capturing these intents enables richer news representations
- The consistency between title and content reflects users' post-read preferences, so incorporating it helps refine news representations
- For cold-start users, leveraging information from similar neighboring users can provide strong support



- Experiments show that intent disentanglement plays an important role
- The method demonstrates strong performance, especially in cold-start scenarios



LANCER : A Lifetime-Aware News Recommender System

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2 College of Information Sciences and Technology, The Pennsylvania State University, USA

LANCER (AAAI 23')

❖ News has a “Lifetime”

- Duration from the birth (initial publish time) to the death (last clicked time)
 - Relatively short (hours, not weeks or months)

❖ News Compete with Other News Within Their Lifetime

- Traditional methods treat all unclicked news as negative samples
- Propose a sampling strategy using the concept of a competition window
 - When two news share an overlapping active period, unclicked item is sampled as a negative example

A Difference in Lifetime

❖ Comparing Click Distributions Between News and Movies

- ❑ News receives concentrated clicks within **6–18 hours** after publication
- ❑ Movies take up to **32 months** to accumulate **80%** of total clicks
 - News reaches **80%** within just **36 hours**

❖ When Lifetime Is Not Considered in Recommendation

- ❑ Previous work tends to recommend many news articles that are already near the end of their lifetime

❖ Recommend News with Sufficient Remaining Lifetime

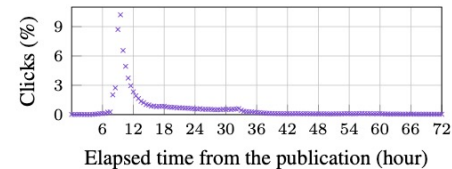


Figure 1: Distribution of average click ratios for news by users over time (Adressa).

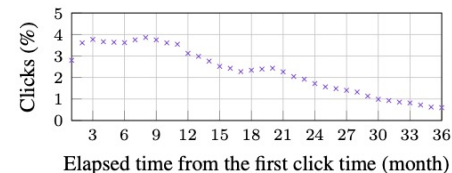


Figure 2: Distribution of average click ratios for items by users over time (Netflix).

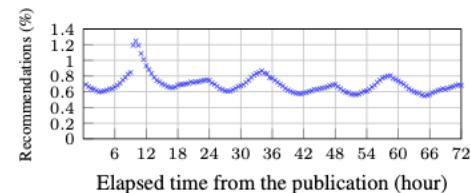


Figure 3: Percentage of news recommended by NRMS (Wu et al. 2019c) by elapsed time from the publication (Adressa).

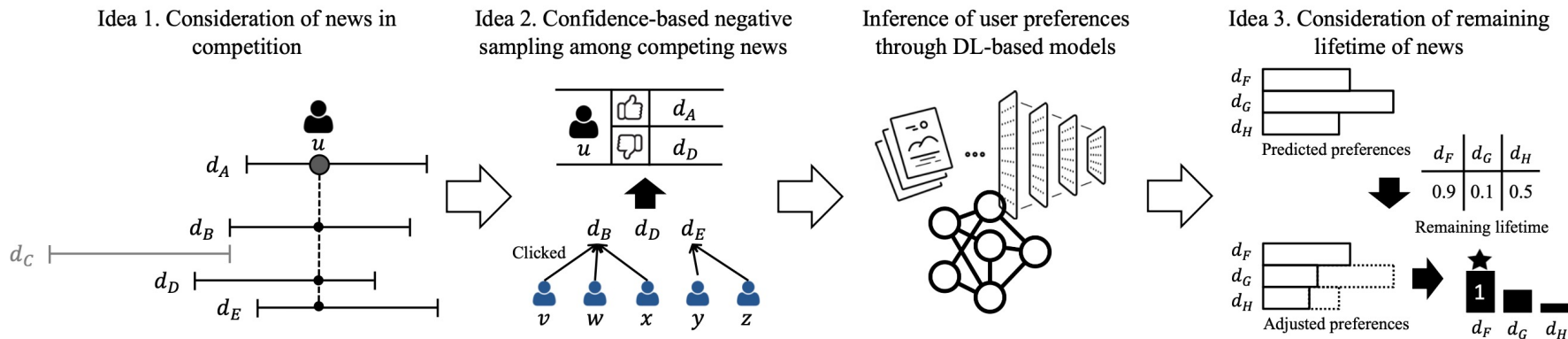


Figure 4: Overview of our proposed approach for news recommendation, *LANCER*.

Negative Sampling Strategy – Previous

Column	Content
Impression ID	91
User ID	U397059
Time	11/15/2019 10:22:32 AM
History	N106403 N71977 N97080 N102132 N97212 N121652
Impressions	N129416-0 N26703-1 N120089-1 N53018-0 N89764-0 N91737-0 N29160-0

❖ Define P/N by Impressions' tag (-0, Negative / -1, Positive)

☐ Positive

- N26703, N120089

☐ Negative

- N129416, N53018, N89764, N01737, N29160

Negative Sampling Strategy – Compete

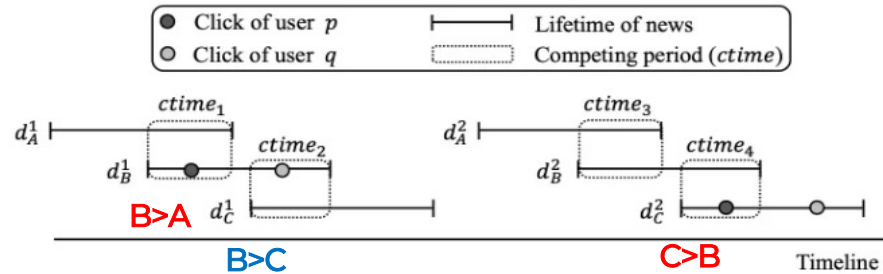


Figure 5: Clicks of two users for six news.

❖ Unclicked news as negative sample in Competition Windows Overlap

□ Clearly separates the preference differences between users p and q

▸ $p: C > B > A$

▸ $q: B > C$

❖ User q 's interest in topic A is actually unknown

□ At the moment q made a choice, news A was not available.

Negative Sampling Strategy - Confidence

- ❖ Only treat news as negative if the user is unlikely to prefer it
- ❖ Leverage Confidence
 - The confidence that “non-clicked news is negative” is defined based on its popularity
 - Low-popularity news is assumed to be *seen but skipped, not unseen*
 - For each selected negative sample,
confidence is computed using the number of other users who clicked competing news

$$conf(u, d_i, d_j) = 1 - \left(\frac{\log(pop(u, d_i, d_j))}{\sum_{d_k \in CPT(d_i)} \log(pop(u, d_i, d_k))} \right),$$

Low popularity
Higher Confidence

$d_j \in CPT(d_i), \quad (3)$

one PS's Negative Set
selected by Compete

Training the DL-based Models

❖ The Concept of News Lifetime Is Independent of the Recommendation Model

- Existing DL-based news recommendation models can be used as-is

$$\mathcal{L}_u = - \sum_{d_i \in \mathcal{I}_u} \log \left(\frac{e^{\hat{p}(u, d_i)}}{e^{\hat{p}(u, d_i)} + \sum_{j=1}^K e^{\hat{p}(u, d_j)}} \right)$$

❖ Model Training with the Proposed Negative Sampling Strategy

- Predicted preference $\hat{p}$ is computed internally using embeddings

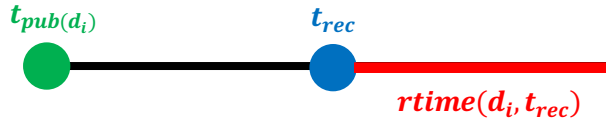
Consideration of Remaining Lifetime

❖ Adjusting Scores When Remaining Lifetime(rtime) is Small

- If remaining lifetime is sufficiently long, the score is mostly preserved
- Setting recommendation points (t_{rec}) appropriately for each experimental environment for the dataset

$$\hat{p}(u, d_i, t_{rec}) = \frac{1}{1 + e^{-\alpha \cdot \underline{rtime(d_i, t_{rec})}}} \cdot \hat{p}(u, d_i)$$

$ltime(d_i)$
(36 hours)



$|ltime(d_i)| - |(t_{rec} - t_{pub(d_i)})|$

Experiment

Datasets	# of users	# of items	# of clicks	Sparsity
Adressa	259,709	24,060	6,067,109	99.9%
MIND	200,000	78,316	4,627,681	99.7%

Table 1: Statistics of two real-world datasets

- ❑ MIND does not provide explicit publish time or click time
- ❑ Use indirect time information
 - **News publish time:** the timestamp of the first impression where the news item appears
 - **News click time:** the impression timestamp in which the news was clicked

Limited Competition Effect

Metric	NRMS				LSTUR				NAML				CNE-SUE			
	Orig (O)	LANCER			Orig (O)	LANCER			Orig (O)	LANCER			Orig* (O)	LANCER		
		C	C/N	C/N/R		C*	C/N	C/N/R		C	C/N	C/N/R		C	C/N	C/N/R
AUC	.551	.620	.637	.663	.571	.587	.603	.617	.632	.687	.715	.740	.600	.638	.644	.657
MRR	.225	.255	.258	.280	.216	.223	.228	.248	.263	.270	.296	.344	.226	.250	.257	.275
G@5	.208	.251	.250	.280	.196	.211	.217	.239	.255	.275	.312	.362	.219	.239	.250	.272
G@10	.273	.329	.333	.363	.283	.293	.301	.321	.334	.364	.392	.435	.295	.322	.336	.355

Table 2: Comparison of accuracy between the original method (*i.e.*, Orig) and variants from *LANCER* for each base model. * denotes $p < 0.0005$ for the paired t -test with $LANCER_{C/N/R}$, respectively (Adressa)

❖ Comparison Between Original (O) and LANCER-C

- ❑ Original : Treats all unclicked news as negative samples
- ❑ The difference about whether LANCER's negative sampling scheme is applied
- ❑ The competition-based negative sampling contributed to performance improvement

Compare with Impression-based

Metric	NRMS				LSTUR				NAML				CNE-SUE			
	Imp	Orig	LCR	Gain	Imp	Orig	LCR	Gain	Imp	Orig	LCR	Gain	Imp	Orig	LCR	Gain
	(I)	(O)	C	vs. I	(I)	(O)	C	vs. I	(I)	(O)	C	vs. I	(I)	(O)	C	vs. I
AUC	.688	.836	.850	23.5.	.587	.663	.688	17.2	.732	.860	.879	20.1	.767	.860	.898	17.1
MRR	.324	.520	.526	62.3	.223	.351	.352	57.8	.399	.552	.562	40.9	.328	.407	.433	32.0
G@5	.325	.560	.567	74.5	.201	.354	.362	80.1	.410	.597	.614	49.8	.542	.690	.741	36.7
G@10	.404	.604	.614	52.0	.287	.414	.424	47.7	.475	.635	.652	37.3	.600	.743	.792	32.0

Table 3: Comparison of accuracy among the impression-based method (*i.e.*, Imp), the original method (*i.e.*, Orig), and $LANCER_C$ (*i.e.*, LCR_C) for each base model, where the Gain (%) indicates the degree of improvement achieved by $LANCER_C$ compared with Imp (MIND)

❖ Comparison with Naive Negative Sampling Within the Same Impression Log

- ❑ All news shown in the same log are treated as competing news
- ❑ Any unselected news in that log becomes a negative sample
- ❑ $LANCER_C$ shows better performance than Imp
 - Ori also shows better performance than Imp

Popularity-based Confidence

Metric	NRMS				LSTUR				NAML				CNE-SUE			
	Orig (O)	LANCER			Orig (O)	LANCER			Orig (O)	LANCER			Orig* (O)	LANCER		
		C	C/N	C/N/R		C*	C/N	C/N/R		C	C/N	C/N/R		C	C/N	C/N/R
AUC	.551	.620	.637	.663	.571	.587	.603	.617	.632	.687	.715	.740	.600	.638	.644	.657
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❖ Improved Performance of LANCER-C/N

- Applying C/N consistently outperformed using C alone across all models

Popularity-based Confidence

Metric	$LANCER_{C/(1-N)}$			
	NRMS	LSTUR	NAML	CNE-SUE
AUC	0.615	0.574	0.665	0.621
MRR	0.242	0.219	0.272	0.244
G@5	0.228	0.197	0.268	0.229
G@10	0.312	0.281	0.350	0.316

Table 4: Accuracy of the variant from $LANCER_{C/(1-N)}$ that is contrary to Idea2 (Adressa)

❖ Sampling High-Popularity News as Negatives with Higher Probability

- ❑ In accuracy metrics: $LANCER_C/N > LANCER_C > LANCER_C/(1-N)$
- ❑ Confirms performance differences based on popularity-aware negative sampling

Considering Remaining Lifetime

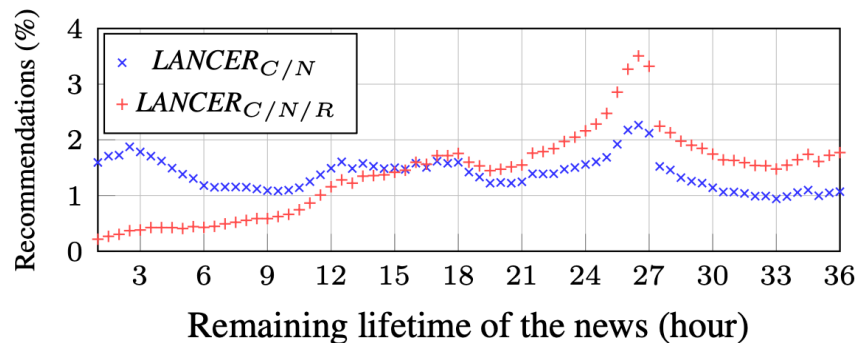


Figure 7: Percentage of news recommended by CNE-SUE by the remaining lifetimes of news (Adressa).

❖ LANCER-C/N/R Avoids Recommending News with Little Remaining Lifetime

- Enables recommendation that considers how much lifetime is left for each news item

Conclusion

- 뉴스의 clicked 패턴을 보면, 뉴스 추천에서 lifetime을 고려하지 않을 수 없다
 - ▶ lifetime이 충분한 뉴스를 고려할 필요가 있다
- 서로 lifetime이 겹치는 뉴스들 간 경쟁에서 negative sample을 뽑을 필요가 있다
 - ▶ confidence를 반영하여, user가 “진짜 선택하지 않은” 뉴스를 sampling

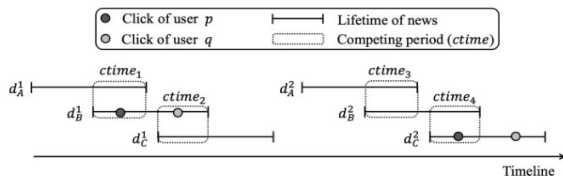


Figure 5: Clicks of two users for six news.

$$conf(u, d_i, d_j) = 1 - \left(\frac{\log(pop(u, d_i, d_j))}{\sum_{d_k \in CPT(d_i)} \log(pop(u, d_i, d_k))} \right),$$

Low popularity
Higher Confidence

$d_j \in CPT(d_i),$ (3)
one PS's Negative Set
selected by Compete

- 실험을 통해 다른 sampling 기법보다 더 좋은 성능을 보임을 확인할 수 있었음
- 또한 remaining lifetime이 충분한 뉴스를 추천하고 있음

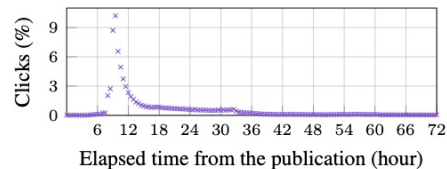


Figure 1: Distribution of average click ratios for news by users over time (Adressa).

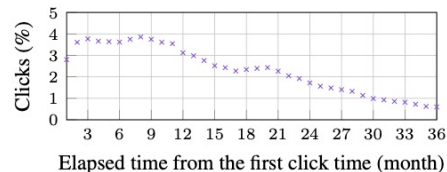


Figure 2: Distribution of average click ratios for items by users over time (Netflix).

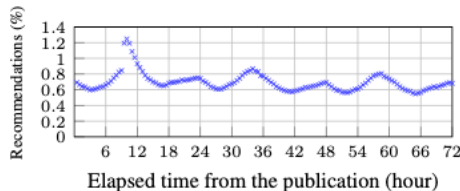


Figure 3: Percentage of news recommended by NRMS (Wu et al. 2019c) by elapsed time from the publication (Adressa).



Is This News Still Interesting to You?: Lifetime-aware Interest Matching for News Recommendation

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Seoul, Korea

LIME (CIKM 25')

❖ Incorporate Temporal Properties into NewsRec

- ❑ Age, Freshness, Lifetime
- ❑ Age serves as indicator of the persistence of user's interest
 - User continues clicking older news, interest may be stable

❖ Lifetime Varies by Topic and by User

- ❑ Previous work assumed fixed lifetime for all news items.
- ❑ In practice, lifetime differs **topic-wise** and **user-wise**

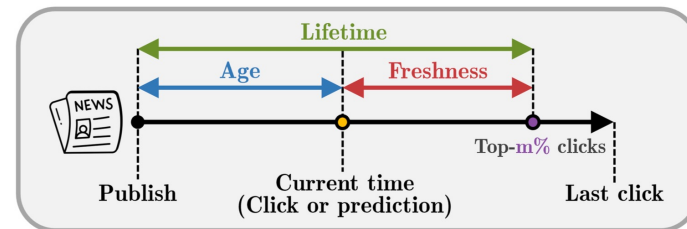


Figure 1: Age, lifetime, and freshness of news.

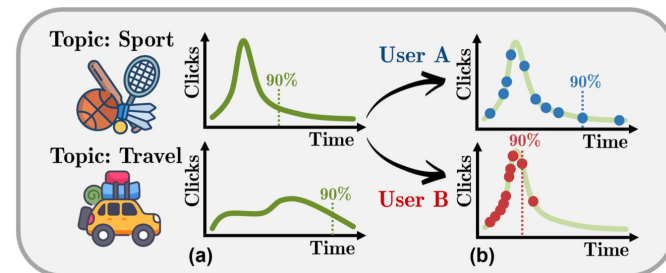
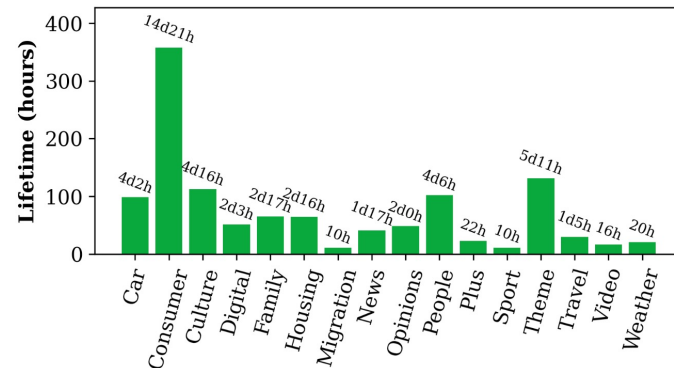


Figure 2: Topic-wise and user-wise news lifetimes.

Analysis of Lifetime

❖ Topic-wise Lifetime

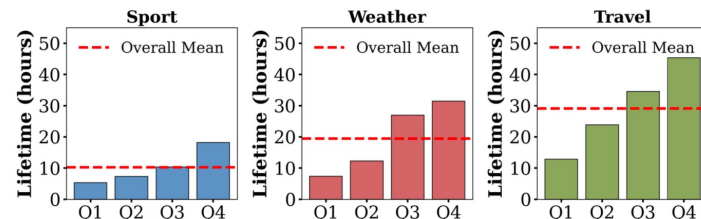
- ❑ Compute average lifetime by grouping news by topic
 - Time taken to reach the top 90% of clicks
 - Significant lifetime differences exist across topics



(a) Topic-wise differences in news lifetime

❖ User-wise Lifetime

- ❑ Divide users into quartiles based on their lifetime distribution and compare average lifetime
- ❑ Average lifetime varies notably across user groups
 - Even for the same topic, users differ in how quickly they lose interest



(b) User-wise differences in news lifetime within topics

Definition of Lifetime

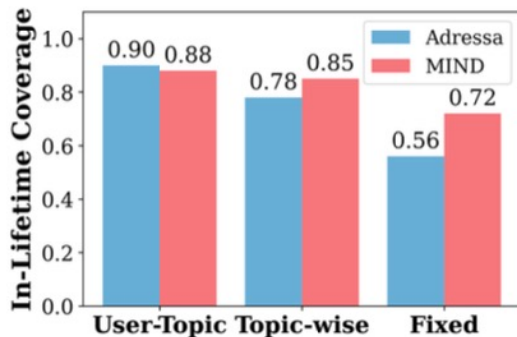
❖ Topic-wise Lifetime

- The lifetime of topic t , denoted $L(t, m)$
- Defined as the average time it takes for the top $m\%$ of clicks to occur across all news within that topic

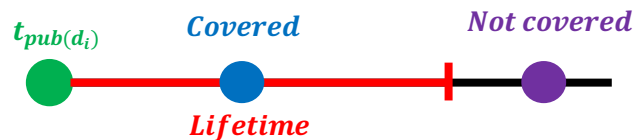
❖ User-Topic Lifetime

- For user u and topic t , the lifetime $L(u, t, m)$
- Defined as the time from the publication of topic- t news until user u reaches the top $m\%$ of clicks for that topic

Importance of Lifetime



(a) Positive click coverage



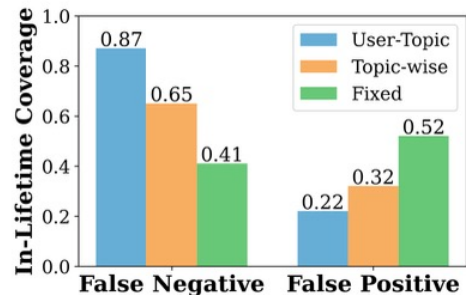
❖ Does the Defined Lifetime Capture the Actual User Click Times?

- ❑ High coverage indicates that the defined lifetime closely reflects the real click distribution
- ❑ Evaluates whether user clicks actually occur within the defined lifetime window

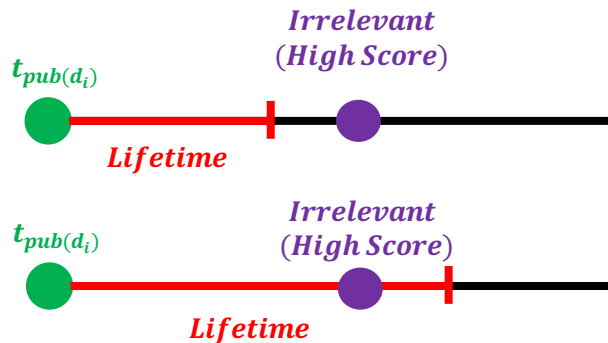
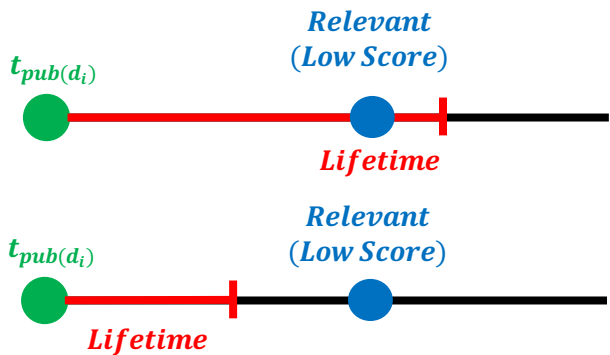
Importance of Lifetime

❖ Ratio of False Negatives and False Positives (Using CROWN)

- ❑ Actually relevant to the user, but model gives low score
 - Proper lifetime would have prevented the news from getting low score
- ❑ Actually irrelevant to the user, but model gives high score
 - Improper Lifetime may contribute to irrelevant news being high score



(b) Prediction error coverage



LIME (Lifetime-aware Interest Matching for nEws recommendation)

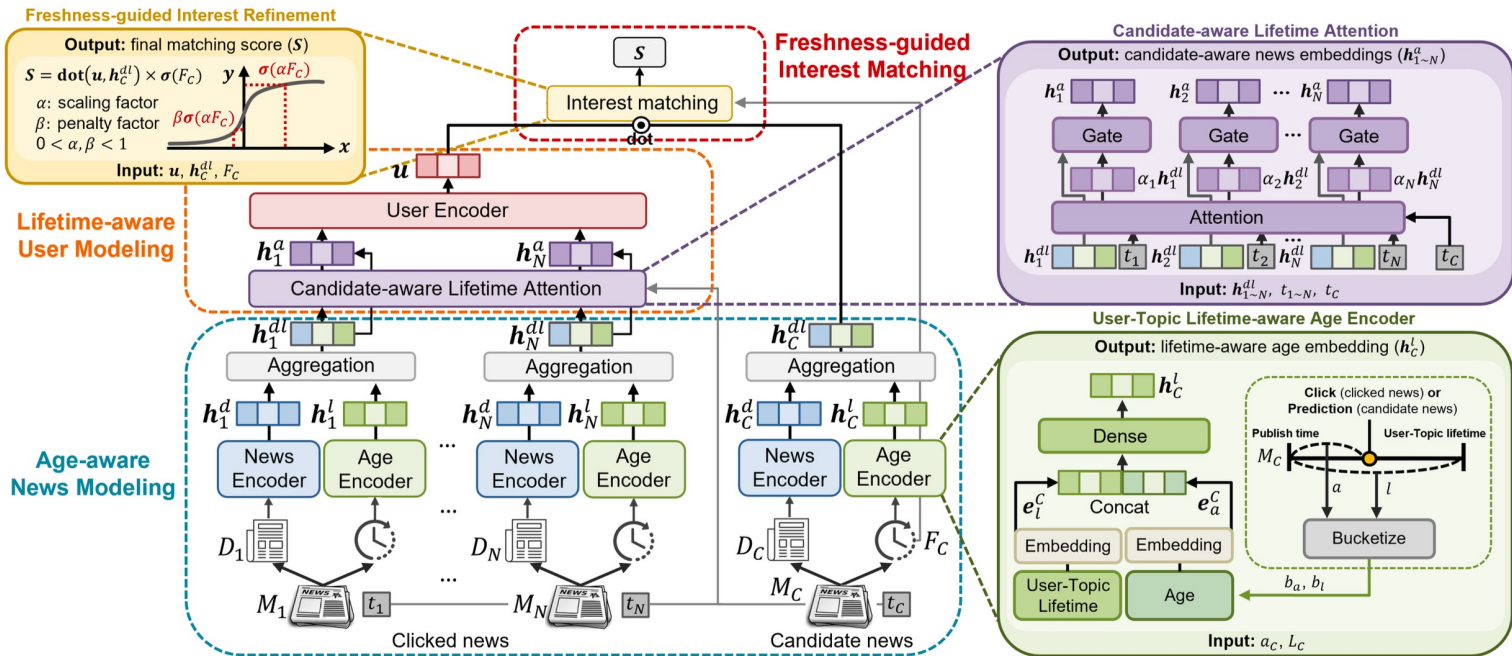


Figure 5: Overview of LIME, which consists of three core components: age-aware news modeling, lifetime-aware user modeling, and freshness-guided interest matching.

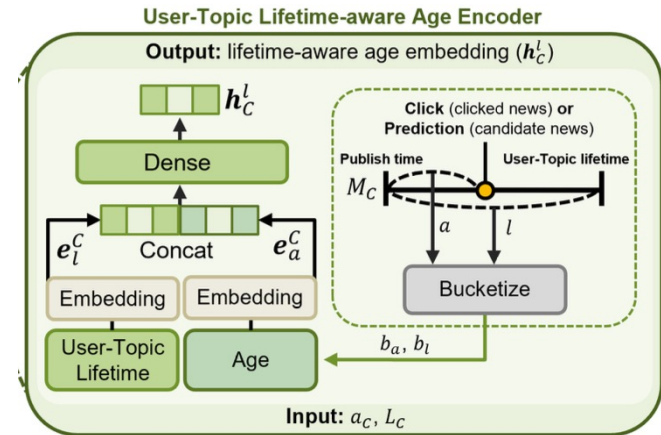
User-Topic Lifetime-aware Age Encoder

❖ Representing Age and Topic Lifetime Using Buckets

- ❑ Convert each bucket ID into an embedding

❖ Re-evaluating Age of News

- ❑ Based on user-specific topic-wise lifetime
- ❑ Determine whether clicked news is relatively young or old compared to the user's topic-wise lifetime



User-Topic Lifetime-aware Age Encoder

$$b = \min \left(\left\lfloor \log(\max(a, 1)) \cdot \frac{B}{\log(\tau)} \right\rfloor, B - 1 \right) \quad b \in \{0, 1, \dots, B - 1\}$$

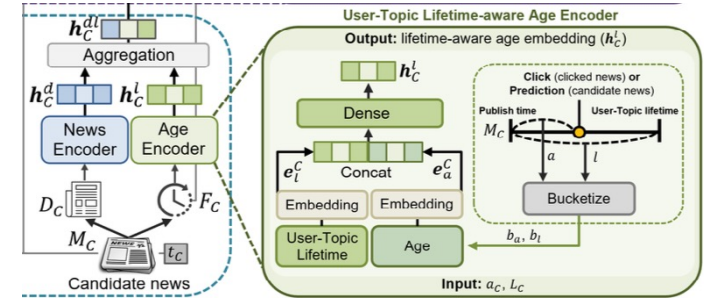
$$\mathbf{h}_n^l = \tanh(\mathbf{W}_n \cdot [\mathbf{e}_a \oplus \mathbf{e}_l] + \mathbf{b}_n) \quad \mathbf{h}_n^{dl} = \text{Concat}(\mathbf{h}_n^d, \mathbf{h}_n^l)$$

❖ News Encoder as Plug-in

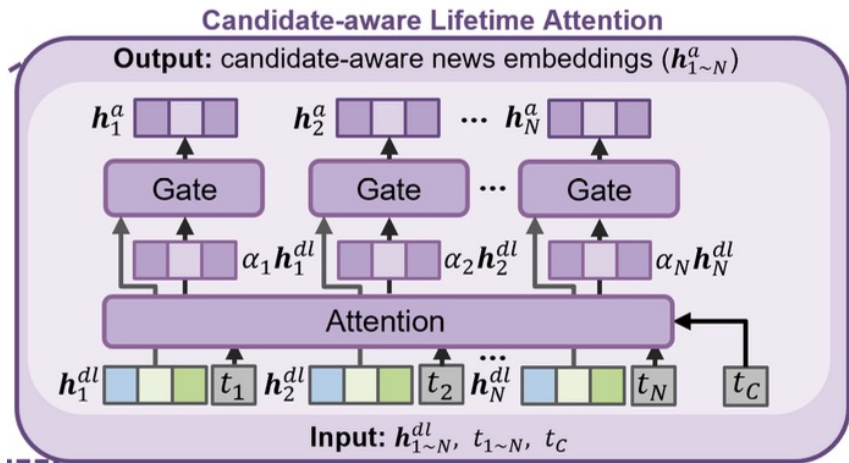
- Generate the news content representation $\mathbf{h}_n^d$

❖ Meaning of the Final Embedding

- For clicked news: indicates how early or late the user clicked at that time
- For candidate news: indicates how old the article is relative to its remaining lifetime at recommendation time



Candidate-aware Lifetime Attention



$$t_i = (c_i \oplus s_i) \cdot W_t + b_t, \quad t_C = (c_C \oplus s_C) \cdot W_t + b_t$$

$$\alpha_i = \frac{\exp(t_C^\top t_i)}{\sum_{j=1}^N \exp(t_C^\top t_j)}$$

$$h_i^a = g_i \odot (\alpha_i \cdot h_i^{dl}) + (1 - g_i) \odot h_i^{dl}, \quad g_i = \sigma(W_g \cdot h_i^{dl})$$

❖ Giving Higher Importance to Clicked News Similar to Candidate

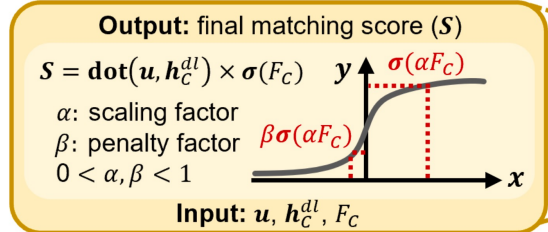
- ❑ Compute importance weights using topic/sub-topic embeddings
- ❑ Reweight each clicked-news embedding using gated residual connection

Freshness-guided Interest Matching

❖ How Fresh Is a Candidate News Article?

- ❑ Relative to the user's Topic-wise interest duration
 - Even for the same candidate news, freshness differs by user
- ❑ Freshness = User-topic lifetime – age of the candidate news
 - Represents the remaining effective lifetime

Freshness-guided Interest Refinement



❖ Incorporating Freshness

- ❑ Freshness < 0 -> the news has exceeded its effective lifetime
 - Apply a penalty
- ❑ Freshness > 0 -> the news still has remaining lifetime
- ❑ Reflected while preserving the original relevance score

$$f(F_c) = \begin{cases} \frac{1}{1 + \exp(-\alpha \cdot F_c)} & \text{if } F_c \geq 0 \\ \beta \cdot \frac{1}{1 + \exp(-\alpha \cdot F_c)} & \text{if } F_c < 0 \end{cases}$$

$$S_{\text{base}} = \mathbf{u}^\top \mathbf{h}_c^{dl} \quad S = S_{\text{base}} \cdot f(F_c)$$

Overall Performance

❖ Integrating LIME into the Existing News Encoder

Table 2: News recommendation accuracy: LIME consistently enhances *all* competitors across *all* metrics in a model-agnostic manner. The improvements are statistically significant under *t*-tests with 95% confidence (p-values < 0.005).

	Metric	NRMS [32]			NAML [29]			LSTUR [1]			CNE-SUE [19]			CROWN [14]		
		Base	LIME	Gain	Base	LIME	Gain	Base	LIME	Gain	Base	LIME	Gain	Base	LIME	Gain
Adressa	AUC	0.7041	0.7933	+12.67%	0.7501	0.8316	+10.87%	0.7605	0.8347	+9.76%	0.7942	0.8627	+8.62%	0.8413	0.8774	+4.29%
	MRR	0.4149	0.5185	+24.97%	0.4344	0.5312	+22.28%	0.5037	0.6025	+19.61%	0.4939	0.6055	+22.59%	0.5526	0.6534	+18.24%
	nDCG@5	0.4046	0.5177	+27.95%	0.4435	0.5466	+23.25%	0.5038	0.6128	+21.64%	0.5196	0.6357	+22.34%	0.5865	0.6823	+16.33%
	nDCG@10	0.4677	0.5835	+24.76%	0.5182	0.6005	+15.88%	0.5564	0.6606	+18.73%	0.5706	0.6736	+18.05%	0.6282	0.7123	+13.39%
MIND	AUC	0.6625	0.6672	+0.71%	0.6631	0.6694	+0.95%	0.6614	0.6675	+0.92%	0.6749	0.6816	+0.99%	0.6821	0.6884	+0.92%
	MRR	0.3060	0.3241	+5.92%	0.3125	0.3282	+5.02%	0.3119	0.3224	+3.37%	0.3212	0.3346	+4.17%	0.3360	0.3422	+1.85%
	nDCG@5	0.3432	0.3564	+3.85%	0.3437	0.3583	+4.25%	0.3428	0.3553	+3.65%	0.3558	0.3683	+3.51%	0.3721	0.3839	+3.17%
	nDCG@10	0.4126	0.4195	+1.67%	0.4106	0.4208	+2.48%	0.4042	0.4189	+3.64%	0.4185	0.4290	+2.51%	0.4314	0.4392	+1.81%

Ablation Study

Table 3: Ablation study of LIME: the proposed strategies (S1–S3) significantly contribute to enhancing the accuracy of LIME. Underline indicates the better result when the same number of strategies are applied. **Bold indicates the best result in all cases.**

		Baseline	LIME									
Metric		NAML	S1	Gain	S3	Gain	S1+S2	Gain	S1+S3	Gain	S1+S2+S3	Gain
Adressa	AUC	0.7501	0.7943	+5.89%	0.7781	+3.73%	0.4716	+7.88%	0.8225	+9.65%	0.8316	+10.87%
	MRR	0.4344	<u>0.4923</u>	+13.33%	0.4716	+8.56%	0.5120	+17.86%	<u>0.5253</u>	+20.93%	0.5312	+22.28%
	nDCG@5	0.4435	<u>0.5020</u>	+13.19%	0.4832	+8.95%	0.5268	+18.78%	<u>0.5384</u>	+21.40%	0.5466	+23.25%
	nDCG@10	0.5182	<u>0.5694</u>	+9.88%	0.5450	+5.17%	0.5860	+13.08%	<u>0.5922</u>	+14.28%	0.6005	+15.88%
MIND	AUC	0.6631	0.6656	+0.38%	0.6652	+0.32%	0.6681	+0.75%	0.6675	+0.66%	0.6694	+0.95%
	MRR	0.3125	<u>0.3187</u>	+1.98%	0.3172	+1.50%	<u>0.3263</u>	+4.42%	0.3241	+3.71%	0.3282	+5.02%
	nDCG@5	0.3437	<u>0.3490</u>	+1.54%	0.3488	+1.48%	<u>0.3545</u>	+3.14%	0.3522	+2.47%	0.3583	+4.25%
	nDCG@10	0.4106	<u>0.4148</u>	+1.02%	0.4140	+0.83%	<u>0.4182</u>	+1.85%	0.4172	+1.61%	0.4208	+2.48%

- ❖ **S1: User-topic Lifetime-aware Age Representation,**
- S2: Candidate-aware lifetime attention, S3: Freshness-guided interest refinement**

- When comparing S1 and S3, S1 showed a larger performance gain
 - Accurately expressing news age based on the user-topic lifetime is crucial

Impact of Lifetime Definition

Table 4: Impact of lifetime definitions across four base models. User-Topic lifetime consistently outperforms Fixed and Topic-wise variants across *all* metrics. Bold and underline indicate the best and second-best results, respectively.

	Metric	NAML	Fixed	Gain	Topic-wise	Gain	User-Topic	Gain	LSTUR	Fixed	Gain	Topic-wise	Gain	User-Topic	Gain
Adressa	AUC	0.7501	0.7768	+3.6%	<u>0.7994</u>	+6.6%	0.8316	+10.9%	0.7605	0.7859	+3.3%	<u>0.7962</u>	+4.7%	0.8347	+9.8%
	MRR	0.4344	0.4525	+4.2%	<u>0.5109</u>	+17.6%	0.5312	+22.3%	0.5037	0.5277	+4.8%	<u>0.5476</u>	+8.7%	0.6025	+19.6%
	nDCG@5	0.4435	0.4643	+4.7%	<u>0.5230</u>	+17.9%	0.5466	+23.3%	0.5038	0.5287	+4.9%	<u>0.5552</u>	+10.2%	0.6128	+21.6%
	nDCG@10	0.5182	0.5317	+2.6%	<u>0.5793</u>	+11.8%	0.6005	+15.9%	0.5564	0.5847	+5.1%	<u>0.6066</u>	+9.0%	0.6606	+18.7%
MIND	AUC	0.6631	0.6636	+0.1%	<u>0.6672</u>	+0.6%	0.6694	+1.0%	0.6614	0.6638	+0.4%	<u>0.6667</u>	+0.8%	0.6675	+0.9%
	MRR	0.3125	0.3109	-0.5%	<u>0.3215</u>	+2.9%	0.3282	+5.0%	0.3119	0.3144	+0.8%	<u>0.3194</u>	+2.4%	0.3224	+3.4%
	nDCG@5	0.3437	0.3382	-1.6%	<u>0.3498</u>	+1.8%	0.3583	+4.3%	0.3428	0.3486	+1.7%	<u>0.3520</u>	+2.7%	0.3553	+3.7%
	nDCG@10	0.4106	0.4091	-0.4%	<u>0.4162</u>	+1.4%	0.4208	+2.5%	0.4042	0.4118	+1.9%	<u>0.4142</u>	+2.5%	0.4189	+3.6%
	Metric	CNE-SUE	Fixed	Gain	Topic-wise	Gain	User-Topic	Gain	CROWN	Fixed	Gain	Topic-wise	Gain	User-Topic	Gain
Adressa	AUC	0.7942	0.8055	+1.4%	<u>0.8328</u>	+4.9%	0.8627	+8.6%	0.8413	0.8468	+0.7%	<u>0.8612</u>	+2.4%	0.8774	+4.3%
	MRR	0.4939	0.5482	+10.9%	<u>0.5825</u>	+17.9%	0.6055	+22.6%	0.5526	0.5742	+3.9%	<u>0.6087</u>	+10.2%	0.6534	+18.2%
	nDCG@5	0.5196	0.5499	+5.8%	<u>0.5928</u>	+14.1%	0.6357	+22.3%	0.5865	0.6061	+3.3%	<u>0.6568</u>	+12.0%	0.6823	+16.3%
	nDCG@10	0.5706	0.6152	+7.8%	<u>0.6406</u>	+12.3%	0.6736	+18.1%	0.6282	0.6485	+3.2%	<u>0.6729</u>	+7.1%	0.7123	+13.4%
MIND	AUC	0.6749	0.6776	+0.4%	<u>0.6790</u>	+0.6%	0.6816	+1.0%	0.6821	0.6831	+0.2%	<u>0.6852</u>	+0.5%	0.6884	+0.9%
	MRR	0.3212	0.3241	+0.9%	<u>0.3293</u>	+2.5%	0.3346	+4.2%	0.3360	0.3392	+1.0%	<u>0.3401</u>	+1.2%	0.3422	+1.9%
	nDCG@5	0.3558	0.3568	+0.3%	<u>0.3628</u>	+2.0%	0.3683	+3.5%	0.3721	0.3748	+0.7%	<u>0.3796</u>	+2.0%	0.3839	+3.2%
	nDCG@10	0.4185	0.4198	+0.3%	<u>0.4226</u>	+1.0%	0.4290	+2.5%	0.4314	0.4342	+0.7%	<u>0.4380</u>	+1.5%	0.4392	+1.8%

❖ Fixed Lifetime (36 hours)

□ A constant 36-hour lifetime was applied

❖ Across all metrics, models using User-Topic Lifetime achieved the highest accuracy

Case Study

Impressed News to U7825

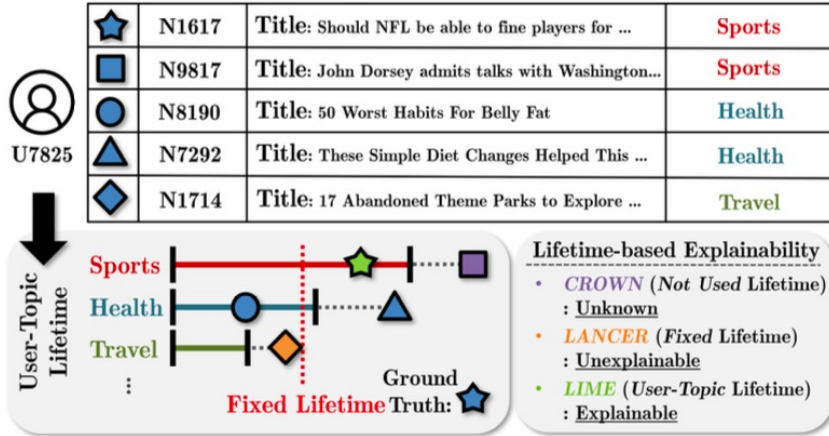
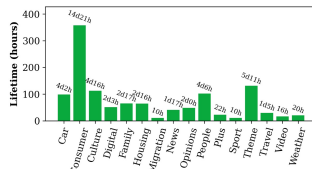


Figure 7: Case study for explainable recommendations from a lifetime perspective.

- U7825 User have different lifetimes for each category
- CROWN
 - Recommendation without Lifetime consideration
- LANCER
 - Defined lifetime is not appropriate, does not reflect actual lifetime
- LIME
 - Because of the definition of appropriate lifetime, able to recommend appropriate news among Impressed News

Conclusion



(a) Topic-wise differences in news lifetime

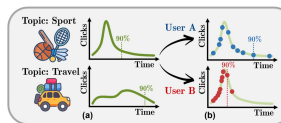
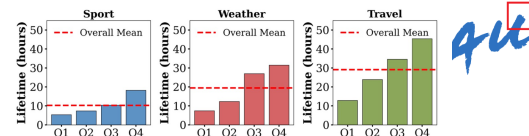
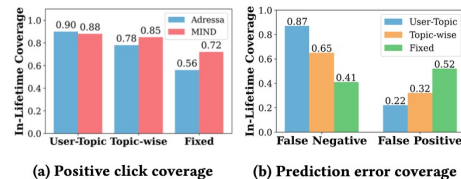


Figure 2: Topic-wise and user-wise news lifetimes.



(b) User-wise differences in news lifetime within topics

- ❑ News has different lifetimes across topics and across users
 - A fixed lifetime cannot properly reflect real news dynamics
- ❑ Proper lifetime design is crucial
 - When comparing model results, lifetime accuracy significantly affects performance
- ❑ Freshness-aware recommendation
 - Remaining effective lifetime is computed based on user-topic lifetime
- ❑ Experiments show that user-topic lifetime provides measurable benefits than other components, and outperforms other lifetime settings



(a) Positive click coverage

(b) Prediction error coverage

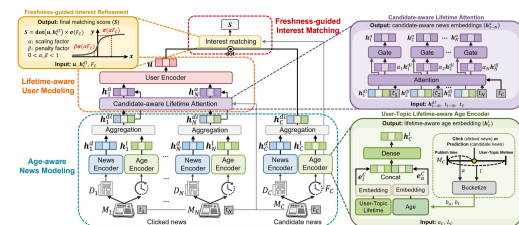


Figure 5: Overview of LIME, which consists of three core components: age-aware news modeling, lifetime-aware user modeling, and freshness-guided interest matching.